

non-rel. Schrödinger equation
or relativistic Dirac equation

LDA or GGA
or hybrids

physical orbitals or not
mesh density and basis set

$$\left(-\frac{\hbar^2}{2m} \vec{\nabla}_{\vec{r}}^2 + v_{\text{ext}}(\vec{r}) + v_H(\vec{r}) + v_{\text{xc}} \right) \varphi_i(\vec{r}) = E_i \varphi_i(\vec{r})$$

pseudopotential

Hartree potential

view EVs as mere Lagrange
multipliers or band structure approx

(ultrasoft/PAW/norm-conserving)
or all-electron

from solving Poisson eq.
or integrating charge density